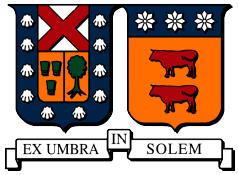
Universidad Técnica Federico Santa María
Departamento de Obras Civiles



Mecánica de Fluidos - CIV244

Unidad IV: Flujo Ideal y Real
Apunte 15: Navier Stokes - Placas Paralelas

(Editado por Vivian Aranda, 2021)

Ecuación de Navier Stokes para flujo compresible

$$\frac{D\vec{u}}{Dt} = \vec{g} - \frac{1}{\rho}\,\nabla p + \frac{\mu}{\rho}\nabla^2\vec{u} + \frac{1}{3}\frac{\mu}{\rho}\nabla(\nabla\cdot\vec{u})$$

$$\frac{\partial\vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u} = \vec{g} - \frac{1}{\rho}\,\nabla p + \frac{\mu}{\rho}\nabla^2\vec{u} + \frac{1}{3}\frac{\mu}{\rho}\nabla(\nabla\cdot\vec{u})$$

Ecuación de Navier Stokes para flujo incompresible

$$(\nabla\cdot\vec{u}) = 0$$

$$\underbrace{\frac{\partial\vec{u}}{\partial t}}_{(1)} + \underbrace{(\vec{u}\cdot\nabla)\vec{u}}_{(2)} = \underbrace{\vec{g}}_{(3)} - \underbrace{\frac{1}{\rho}\,\nabla p}_{(4)} + \underbrace{\frac{\mu}{\rho}\nabla^2\vec{u}}_{(5)}$$

(1) $\dfrac{\partial\vec{u}}{\partial t}$ Representa la aceleración local de una partícula de fluido en un punto fijo, del espacio. Para flujo permanente, $\dfrac{\partial\vec{u}}{\partial t} = 0$.

(2) $(\vec{u}\cdot\nabla)\vec{u}$ Aceleración convectiva de la partícula de fluido, y predice como el flujo difiere de un lugar espacial al siguiente en el mismo instante de tiempo. Flujo uniforme $\Longrightarrow (\vec{u}\cdot\nabla)\vec{u} = 0$.

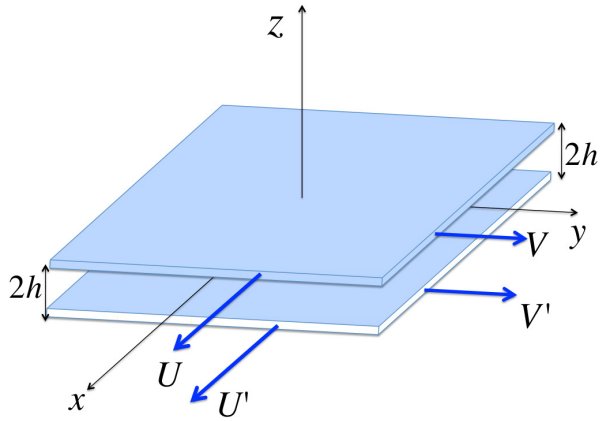
(3) $\vec{g}$ Aceleración debido a la gravedad.

(4) $-\dfrac{1}{\rho}\,\nabla p$ Aceleración de presión.

(5) $\dfrac{\mu}{\rho}\nabla^2\vec{u}$ Desaceleración viscosa.

Flujo entre dos placas paralelas infinitas

Considere un flujo incompresible ($\nabla \cdot \vec{u} = 0$), permanente entre dos placas paralelas infinitas, como muestra la figura.



La placa superior está a la altura $z = h$ y la placa inferior está a la altura $z = -h$. La velocidad de la placa superior es U en la dirección x y V en la dirección y, por otro lado, la velocidad de la placa inferior es $U^{'}$ en la dirección x y $V^{'}$ en la dirección y. En este caso, $\partial p/\partial y = 0$ y buscamos encontrar el vector $\vec{u} = (u(z), v(z), 0)$ y la presión p.

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + w\,\frac{\partial u}{\partial z} = g_x - \frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$

$$\frac{\partial v}{\partial t} + u\,\frac{\partial v}{\partial x} + v\,\frac{\partial v}{\partial y} + w\,\frac{\partial v}{\partial z} = g_y - \frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right)$$

$$\frac{\partial w}{\partial t} + u\,\frac{\partial w}{\partial x} + v\,\frac{\partial w}{\partial y} + w\,\frac{\partial w}{\partial z} = g_z - \frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right)$$

$$\text{permanente} \quad \Longrightarrow \quad \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = \frac{\partial w}{\partial t} = 0$$

$$w = 0 \quad \rightarrow \quad \frac{\partial w}{\partial x} = 0, \quad \frac{\partial w}{\partial y} = 0, \quad \frac{\partial w}{\partial z} = 0,$$

$$\text{como} \quad u = u(z) \neq 0 \quad \Longrightarrow \quad \frac{\partial u}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = 0,$$

$$\text{como} \quad v = v(z) \neq 0 \quad \Longrightarrow \quad \frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 0,$$

$$\underbrace{\frac{\partial u}{\partial t}}_{0} + u\ \underbrace{\frac{\partial u}{\partial x}}_{0} + v\ \underbrace{\frac{\partial u}{\partial y}}_{0} + \underbrace{w}_{0}\ \frac{\partial u}{\partial z} = \underbrace{g_x}_{0} - \frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\underbrace{\frac{\partial^2 u}{\partial x^2}}_{0} + \underbrace{\frac{\partial^2 u}{\partial y^2}}_{0} + \frac{\partial^2 u}{\partial z^2}\right)$$

$$\underbrace{\frac{\partial v}{\partial t}}_{0} + u\ \underbrace{\frac{\partial v}{\partial x}}_{0} + v\ \underbrace{\frac{\partial v}{\partial y}}_{0} + \underbrace{w}_{0}\ \frac{\partial v}{\partial z} = \underbrace{g_y}_{0} - \underbrace{\frac{1}{\rho}\frac{\partial p}{\partial y}}_{0} + \nu\left(\underbrace{\frac{\partial^2 v}{\partial x^2}}_{0} + \underbrace{\frac{\partial^2 v}{\partial y^2}}_{0} + \frac{\partial^2 v}{\partial z^2}\right)$$

$$\underbrace{\frac{\partial w}{\partial t}}_{0} + u\ \underbrace{\frac{\partial w}{\partial x}}_{0} + v\ \underbrace{\frac{\partial w}{\partial y}}_{0} + \underbrace{w}_{0}\ \underbrace{\frac{\partial w}{\partial z}}_{0} = \underbrace{g_z}_{-g} - \frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\left(\underbrace{\frac{\partial^2 w}{\partial x^2}}_{0} + \underbrace{\frac{\partial^2 w}{\partial y^2}}_{0} + \underbrace{\frac{\partial^2 w}{\partial z^2}}_{0}\right)$$

$$(1) \qquad 0 = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\frac{\partial^2 u}{\partial z^2}$$

$$(2) \qquad 0 = \nu\frac{\partial^2 v}{\partial z^2}$$

$$(3) \qquad 0 = -g - \frac{1}{\rho}\frac{\partial p}{\partial z}$$

flujo incompresible $\implies$ $(\nabla \cdot \vec{u}) = \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} + \dfrac{\partial w}{\partial z} = 0$

Como, $\dfrac{\partial p}{\partial y} = 0 \implies \displaystyle\int dp = \int 0\ dy$

$$p(x,z) = p_o + \varphi(x,z)$$

de la ecuación (3)

$$\begin{aligned}
\frac{\partial p(x,z)}{\partial z} &= -g\rho \\
0 + \frac{\partial \varphi(x,z)}{\partial z} &= -g\rho \\
\int d\varphi &= \int -g\rho \; dz \\
\varphi(x,z) &= -g\rho z + \psi(x)
\end{aligned}$$

$$\Longrightarrow \quad p(x,z) = p_o + \varphi(x,z) = p_o - g\rho z + \psi(x)$$

de la ecuación (1)

$$\begin{aligned}
\frac{\mu}{\rho}\frac{\partial^2 u}{\partial z^2} &= \frac{1}{\rho}\frac{\partial p}{\partial x} \\
\frac{\partial}{\partial z}\left(\frac{\partial u}{\partial z}\right) &= \frac{1}{\mu}\frac{\partial \psi(x)}{\partial x} \\
\int d\left(\frac{du}{dz}\right) &= \int \frac{1}{\mu}\frac{\partial \psi(x)}{\partial x}\; dz \\
\frac{du}{dz} &= \frac{1}{\mu}\frac{\partial \psi(x)}{\partial x} z + C_1 \\
\int du &= \int \left[\frac{1}{\mu}\frac{\partial \psi(x)}{\partial x} z + C_1\right] dz \\
u &= \frac{1}{\mu}\frac{\partial \psi(x)}{\partial x}\frac{z^2}{2} + C_1 \; z + C_2
\end{aligned}$$

recordar que u es solo función de z, por lo tanto, el término $\dfrac{\partial \psi(x)}{\partial x}$ es necesariamente una constante, es decir:

$$\frac{\partial \psi(x)}{\partial x} = G \quad \Longrightarrow \quad \psi(x) = G \; x \qquad \Longrightarrow \qquad p(x,z) = p_o - g\rho z + G \; x$$

por lo tanto,

$$u(z) = \frac{G}{2\mu} \; z^2 + C_1 \; z + C_2$$

condiciones de borde: $\quad u(z = h) = U \quad$ y $\quad u(z = -h) = U^{'}$

$$u(h) = U = \frac{G}{2\mu}\, h^2 + C_1\, h + C_2$$

$$u(-h) = U' = \frac{G}{2\mu}\, (-h)^2 - C_1\, h + C_2$$

sumando $\Longrightarrow$ $C_2 = \dfrac{U + U'}{2} - \dfrac{Gh^2}{2\mu}$ y restando $\Longrightarrow$ $C_1 = \dfrac{U - U'}{2h}$

$$u(z) = \frac{G}{2\mu}\, z^2 + \frac{U - U'}{2h}\, z + \frac{U + U'}{2} - \frac{Gh^2}{2\mu}$$

de la ecuación (2)

$$\frac{\partial^2 v}{\partial z^2} = 0$$

$$\frac{\partial}{\partial z}\left(\frac{\partial u}{\partial z}\right) = 0$$

$$\int d\left(\frac{dv}{dz}\right) = \int 0\; dz$$

$$\frac{dv}{dz} = A_1$$

$$\int dv = \int A_1\; dz$$

$$v(z) = A_1\, z + A_2$$

condiciones de borde: $v(z = h) = V$ y $v(z = -h) = V'$

$$v(h) = V = A_1\, h + A_2$$

$$v(-h) = V' = A_1\, (-h) + A_2$$

sumando $\Longrightarrow$ $A_2 = \dfrac{V + V'}{2}$ y restando $\Longrightarrow$ $A_1 = \dfrac{V - V'}{2h}$

$$v(z) = \frac{V - V'}{2h}\, z + \frac{V + V'}{2}$$

Finalmente para el ejemplo de placas paralelas:

$$
\begin{aligned}
u(z) &= \frac{G}{2\mu}\,(z^2-h^2)+\frac{U-U^{'}}{2h}\,z+\frac{U+U^{'}}{2}\\
v(z) &= \frac{V-V^{'}}{2h}\,z+\frac{V+V^{'}}{2}\\
w &= 0\\
p(x,z) &= p_o-g\rho z+G\,x
\end{aligned}
$$

Tensor de esfuerzo para fluido Newtoniano

Tensor de deformación pura :

$$
\tilde{\tilde{e}} = \underbrace{\begin{pmatrix} \frac{\partial u}{\partial x} & 0 & 0\\ 0 & \frac{\partial v}{\partial y} & 0\\ 0 & 0 & \frac{\partial w}{\partial z}\end{pmatrix}}_{deformacion\ volumetrica} + \underbrace{\frac{1}{2}\begin{pmatrix} 0 & \frac{\partial u}{\partial y}+\frac{\partial v}{\partial x} & \frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\\ \frac{\partial v}{\partial x}+\frac{\partial u}{\partial y} & 0 & \frac{\partial v}{\partial z}+\frac{\partial w}{\partial y}\\ \frac{\partial w}{\partial x}+\frac{\partial u}{\partial z} & \frac{\partial w}{\partial y}+\frac{\partial v}{\partial z} & 0\end{pmatrix}}_{deformacion\ angular}
$$

$$
= \frac{1}{2}\begin{pmatrix} \frac{\partial u}{\partial x}+\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}+\frac{\partial v}{\partial x} & \frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}\\ \frac{\partial v}{\partial x}+\frac{\partial u}{\partial y} & \frac{\partial v}{\partial y}+\frac{\partial v}{\partial y} & \frac{\partial v}{\partial z}+\frac{\partial w}{\partial y}\\ \frac{\partial w}{\partial x}+\frac{\partial u}{\partial z} & \frac{\partial w}{\partial y}+\frac{\partial v}{\partial z} & \frac{\partial w}{\partial z}+\frac{\partial w}{\partial z}\end{pmatrix}
$$

El **tensor de esfuerzos de un fluido Newtoniano** en coordenadas (x,y,z) está dado por:

$$
\tilde{\tilde{\sigma}} = -p\;\tilde{\tilde{I}} + 2\mu\;\tilde{\tilde{e}} - \frac{2}{3}\mu(\nabla\cdot\vec{u})\;\tilde{\tilde{I}}
$$

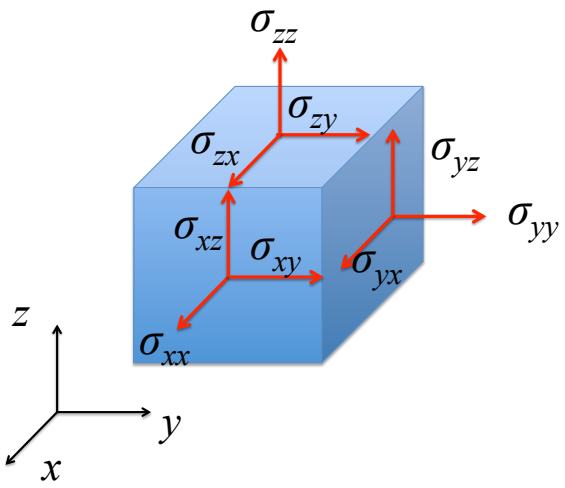
$$
\sigma_{ij} = -p\;\delta_{ij} + 2\mu\;e_{ij} - \frac{2}{3}\mu(\nabla\cdot\vec{u})\;\delta_{ij}
$$

Con e_{ij} el tensor de deformación pura, que incluye la deformación volumétrica mas la deformación angular. Por lo tanto los términos de σ_{ij} corresponden a:

$-p\ \delta_{ij}$: presión termodinámica

$2\mu\ e_{ij}$: esfuerzo asociado a deformación volumétrica y deformación angular

$-\frac{2}{3}\mu(\nabla \cdot \vec{u})\ \delta_{ij}$: esfuerzo que ocurre sólo en flujos compresibles (dilatación viscosa)



En el caso de **fluido incompresible**, se tiene que $(\nabla \cdot \vec{u}) = 0$, por lo tanto el tensor de esfuerzos de un fluido Newtoniano se reduce a: $\tilde{\tilde{\sigma}} = -p\ \tilde{\tilde{I}} + 2\mu\ \tilde{\tilde{e}}$

$$\sigma_{ij} = -p\ \delta_{ij} + 2\mu\ e_{ij}$$

$$= -p\ \delta_{ij} + \underbrace{\qquad\qquad}_{\tau_{ij}}$$

esfuerzo asociado a deformación volumétrica y angular

$$= \begin{pmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{pmatrix} + 2\ \mu\ \frac{1}{2} \begin{pmatrix} \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial x}\right) & \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) & \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \\ \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right) & \left(\frac{\partial v}{\partial y} + \frac{\partial v}{\partial y}\right) & \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) \\ \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right) & \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right) & \left(\frac{\partial w}{\partial z} + \frac{\partial w}{\partial z}\right) \end{pmatrix}$$

τ_{ij} con $i = j$, corresponde a esfuerzo asociado a deformación volumétrica
τ_{ij} con $i \neq j$, corresponde a esfuerzo asociado a deformación angular

Ejemplo: El esfuerzo $\sigma_{zx} = \tau_{zx}$ corresponde a,

$$\begin{aligned}
\sigma_{zx} &= 0 + \tau_{zx} = \tau_{zx} \\
&= \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\
&= \mu \left(\frac{\partial u}{\partial z} + \underbrace{\frac{\partial w}{\partial x}}_{0} \right) \\
&= \mu \frac{\partial}{\partial z} \left(\frac{G}{2\mu} \, (z^2 - h^2) + \frac{U - U^{'}}{2h} \, z + \frac{U + U^{'}}{2} \right) \\
&= \mu \left(\frac{G}{2\mu} \, 2z + \frac{U - U^{'}}{2h} \right) = Gz + \mu \frac{U - U^{'}}{2h}
\end{aligned}$$

Ejemplo: El esfuerzo σ_{xx} corresponde a,

$$\begin{aligned}
\sigma_{xx} &= -p + \tau_{xx} \\
&= -p(x,z) + \mu \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \right) \\
&= -p_o + g\rho z - G \, x + 2\mu \underbrace{\frac{\partial}{\partial x} \left(\frac{G}{2\mu} \, (z^2 - h^2) + \frac{U - U^{'}}{2h} \, z + \frac{U + U^{'}}{2} \right)}_{0}
\end{aligned}$$

Tensor de rotación de cuerpo rígido

$$\tilde{\tilde{\xi}} = \frac{1}{2} \begin{pmatrix} 0 & \frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} & \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} & 0 & \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \\ \frac{\partial w}{\partial x} - \frac{\partial u}{\partial z} & \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 & \omega_z & -\omega_y \\ -\omega_z & 0 & \omega_x \\ \omega_y & -\omega_x & 0 \end{pmatrix}$$

donde:

$$\nabla \times \vec{u} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \, \hat{\imath} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \, \hat{\jmath} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, \hat{k} = \omega_x \, \hat{\imath} + \omega_y \, \hat{\jmath} + \omega_z \, \hat{k} = (\omega_x, \omega_y, \omega_z) = \vec{\omega}$$

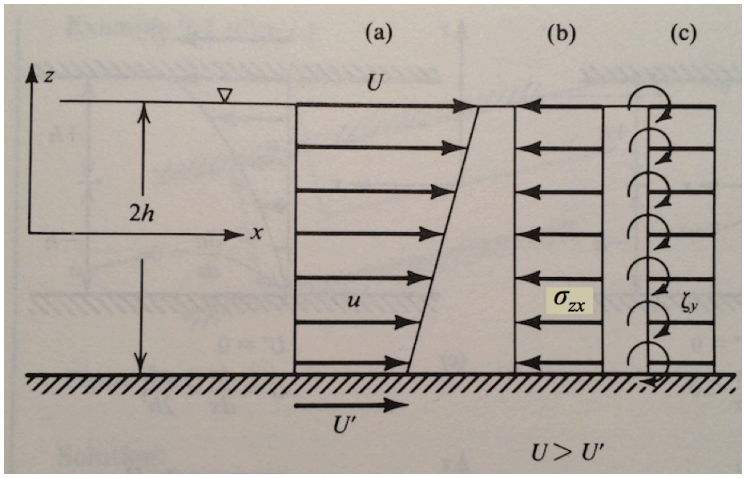
$\vec{\omega}$: vector vorticidad.

Ejemplo: La vorticidad ω_y en la dirección y corresponde a,

$$\begin{aligned}
\omega_y &= \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) \\
&= \left(\frac{\partial u}{\partial z} - \underbrace{\frac{\partial w}{\partial x}}_{0}\right) \\
&= \frac{\partial}{\partial z}\left(\frac{G}{2\mu}\,(z^2 - h^2) + \frac{U - U^{'}}{2h}\,z + \frac{U + U^{'}}{2}\right) \\
&= \left(\frac{G}{2\mu}\,2z + \frac{U - U^{'}}{2h}\right) \\
&= \frac{G}{\mu}\,z + \frac{U - U^{'}}{2h}
\end{aligned}$$

Flujo de Couette

Considerando un flujo bidimensional en el plano (x, z), en este caso la placa inferior se mueve en la dirección x a una velocidad $U^{'}$, y la parte superior se encuentra a presión atmosférica y se mueve a velocidad U, donde $U > U^{'}$ y además $G = dp/dx = 0$.



(a) Velocidad: $u(z) = \dfrac{U - U^{'}}{2h}\, z + \dfrac{U + U^{'}}{2}$ (ecuación lineal)

(b) Esfuerzo: $\sigma_{zx} = \mu \dfrac{U - U^{'}}{2h} = cte$

(c) Vorticidad: $\omega_y = \dfrac{U - U^{'}}{2h} = cte$

Velocidad:

$$\begin{aligned} u(z) &= \frac{G}{2\mu}\,(z^2 - h^2) + \frac{U - U^{'}}{2h}\, z + \frac{U + U^{'}}{2} = \frac{0}{2\mu}\,(z^2 - h^2) + \frac{U - U^{'}}{2h}\, z + \frac{U + U^{'}}{2} \\ &= \underbrace{\frac{U - U^{'}}{2h}}_{>0}\, z + \underbrace{\frac{U + U^{'}}{2}}_{>o} \end{aligned}$$

$$\begin{aligned} u(h) &= \frac{U - U^{'}}{2h}\, h + \frac{U + U^{'}}{2} = U \\ u(-h) &= -\frac{U - U^{'}}{2h}\, h + \frac{U + U^{'}}{2} = U^{'} \end{aligned}$$

Presión:

$$p(x,z) = p_o - g\rho z + 0 \cdot x = p_o - g\rho z, \qquad \text{donde} \qquad \frac{dp}{dx} = 0$$

En este caso particular, $p(x,h) = p_{atm}$,

$$\begin{aligned} p(x,h) &= p_o - g\rho h \\ p_{atm} &= p_o - g\rho h \\ p_o &= p_{atm} + g\rho h \end{aligned}$$

por lo tanto,

$$p(x,z) = p_{atm} + g\rho(h - z)$$

Esfuerzo σ_{zx}:

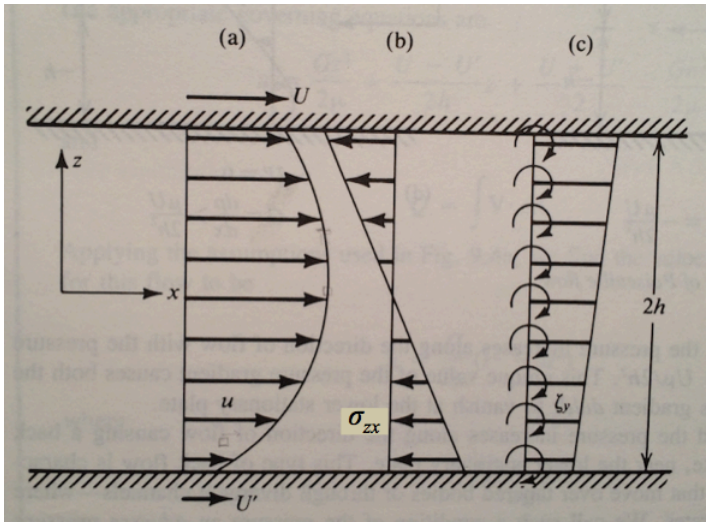
$$\sigma_{zx} = Gz + \mu\frac{U - U'}{2h} = 0 \; z + \mu\frac{U - U'}{2h} = \underbrace{\mu\frac{U - U'}{2h}}_{>0} = cte$$

Vorticidad en la dirección y:

$$\omega_y = \frac{G}{\mu}\, z + \frac{U - U'}{2h} = \frac{0}{\mu}\, z + \frac{U - U'}{2h} = \underbrace{\frac{U - U'}{2h}}_{>0} = cte$$

Flujo de Poiseuille

Considerando un flujo bidimensional en el plano (x, z), en este caso la placa inferior se mueve en la dirección x a una velocidad $U^{'}$, y la placa superior se mueve a velocidad U, donde $U > U^{'} > 0$ y además $G = dp/dx \neq 0$.



(a) Velocidad: $u(z) = \dfrac{G}{2\mu}\,(z^2 - h^2) + \dfrac{U - U^{'}}{2h}\,z + \dfrac{U + U^{'}}{2}$

(b) Esfuerzo: $\sigma_{zx} = Gz + \mu\dfrac{U - U^{'}}{2h}$

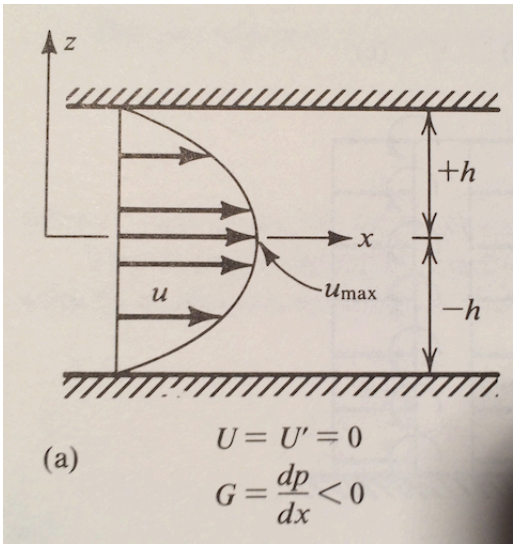
(c) Vorticidad: $\omega_y = \dfrac{G}{\mu}\,z + \dfrac{U - U^{'}}{2h}$

Notar que el perfil de velocidades en este caso es de tipo parabólico, debido a la existencia de un gradiente de presiones.

Tipos de Flujo de Poiseuille

Caso (a)

Donde $U = U' = 0$ y $G = \dfrac{dp}{dx} < 0$



(a) Velocidad: $u(z) = \underbrace{G}_{<0} \dfrac{1}{2\mu}\,(z^2 - h^2) + \underbrace{\dfrac{U - U'}{2h}}_{0}\, z + \underbrace{\dfrac{U + U'}{2}}_{0} = \underbrace{G}_{<0} \dfrac{1}{2\mu}\,(z^2 - h^2)$

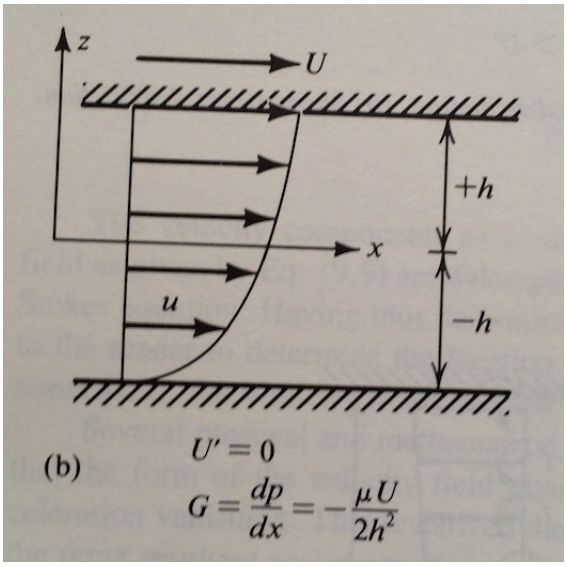
$$
\begin{aligned}
u(h) &= \underbrace{G}_{<0} \frac{1}{2\mu}\,(h^2 - h^2) = 0 \\
u(0) &= \underbrace{G}_{<0} \frac{1}{2\mu}\,(0^2 - h^2) = \underbrace{G}_{<0} \frac{1}{2\mu}\,(-h^2) > 0 \\
u(-h) &= \underbrace{G}_{<0} \frac{1}{2\mu}\,((-h)^2 - h^2) = 0
\end{aligned}
$$

(b) Esfuerzo: $\sigma_{zx} = \underbrace{G}_{<0} z$

(c) Vorticidad: $\omega_y = \underbrace{G}_{<0} \dfrac{1}{\mu}\, z$

Caso (b)

Donde $U^{'} = 0$ y $G = \dfrac{dp}{dx} = -\left(\dfrac{\mu U}{2h^2}\right)$



(a) Velocidad:

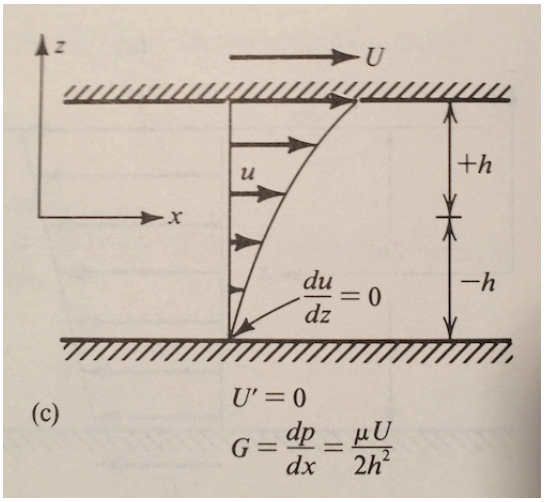
$$\begin{aligned} u(z) &= G\frac{1}{2\mu}\,(z^2 - h^2) + \frac{U - U^{'}}{2h}\,z + \frac{U + U^{'}}{2} \\ &= -\left(\frac{\mu U}{2h^2}\right)\frac{1}{2\mu}\,(z^2 - h^2) + \frac{U}{2h}\,z + \frac{U}{2} \end{aligned}$$

(b) Esfuerzo: $\sigma_{zx} = Gz + \mu\dfrac{U - U^{'}}{2h} = -\left(\dfrac{\mu U}{2h^2}\right)z + \mu\dfrac{U}{2h}$

(c) Vorticidad: $\omega_y = \dfrac{G}{\mu}\,z + \dfrac{U - U^{'}}{2h} = -\left(\dfrac{\mu U}{2h^2}\right)\dfrac{1}{\mu}\,z + \dfrac{U}{2h}$

Caso (c)

Donde $U^{'} = 0$ y $G = \dfrac{dp}{dx} = \left(\dfrac{\mu U}{2h^2}\right)$



(a) Velocidad:

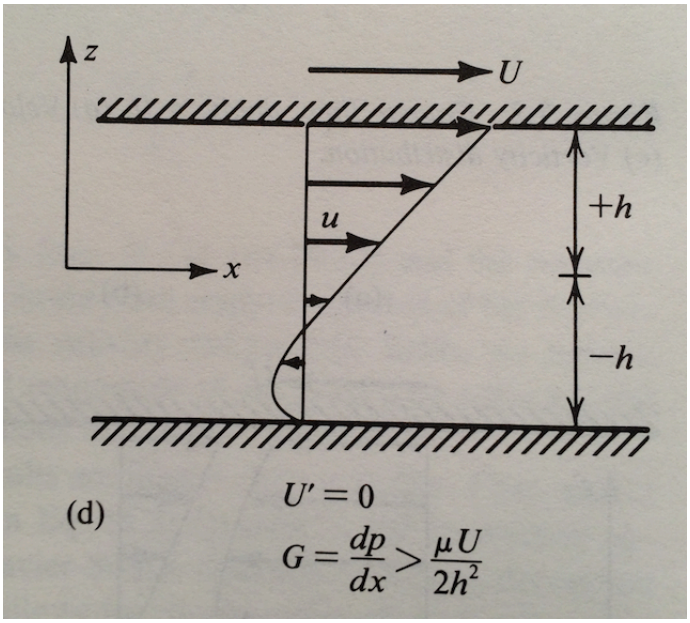
$$\begin{aligned} u(z) &= G\frac{1}{2\mu}\,(z^2 - h^2) + \frac{U - U^{'}}{2h}\,z + \frac{U + U^{'}}{2} \\ &= \left(\frac{\mu U}{2h^2}\right)\frac{1}{2\mu}\,(z^2 - h^2) + \frac{U}{2h}\,z + \frac{U}{2} \end{aligned}$$

(b) Esfuerzo: $\sigma_{zx} = Gz + \mu\dfrac{U - U^{'}}{2h} = \left(\dfrac{\mu U}{2h^2}\right) z + \mu\dfrac{U}{2h}$

(c) Vorticidad: $\omega_y = \dfrac{G}{\mu}\,z + \dfrac{U - U^{'}}{2h} = \left(\dfrac{\mu U}{2h^2}\right)\dfrac{1}{\mu}\,z + \dfrac{U}{2h}$

Caso (d)

Donde $U' = 0$ y $G = \dfrac{dp}{dx} > \left(\dfrac{\mu U}{2h^2}\right)$



(a) Velocidad:

$$\begin{aligned} u(z) &= G\frac{1}{2\mu}\,(z^2 - h^2) + \frac{U - U'}{2h}\,z + \frac{U + U'}{2} \\ &= \underbrace{G}_{>(\mu U/2h^2)}\,\frac{1}{2\mu}\,(z^2 - h^2) + \frac{U}{2h}\,z + \frac{U}{2} \end{aligned}$$

(b) Esfuerzo: $\sigma_{zx} = Gz + \mu\dfrac{U - U'}{2h} = \underbrace{G}_{>(\mu U/2h^2)}\,z + \mu\dfrac{U}{2h}$

(c) Vorticidad: $\omega_y = G\dfrac{1}{\mu}\,z + \dfrac{U - U'}{2h} = \underbrace{G}_{>(\mu U/2h^2)}\,\dfrac{1}{\mu}\,z + \dfrac{U}{2h}$

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